

Circular Palindromes

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Problem

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A palindrome is a string that reads the same from left to right as it does from right to left.

Given a string, S , of N lowercase English letters, we define a k -length rotation as cutting the first k characters from the beginning of S and appending them to the end of S . For each S , there are N possible k -length rotations (where $0 \leq k < N$). See the Explanation section for examples.

Given N and S , find all N k -length rotations of S ; for each rotated string, S_k , print the maximum possible length of any palindromic substring of S_k on a new line.

Input Format

The first line contains an integer, N (the length of S).

The second line contains a single string, S .

Constraints

- $1 \leq N \leq 5 \times 10^5$
- $0 \leq k < N$
- S is comprised of lowercase English letters.

Output Format

There should be N lines of output, where each line k contains an integer denoting the maximum length of any palindromic substring of rotation S_k .

Sample Input 0

```
13
aaaaabbbbbaaaa
```

Sample Output 0

Author: shaka_shadows
Difficulty: Advanced
Max Score: 120
Submitted By: 13760

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```
1 import java.io.*;
2 import java.util.*;
3 import static java.util.stream.Collectors.joining;
4
5 class Result {
6
7     /*
8      * Complete the 'circularPalindromes' function below.
9      *
10     * The function is expected to return an INTEGER_ARRAY.
11     * The function accepts STRING s as parameter.
12     */
13
14     public static List<Integer> circularPalindromes(String s) {
15
16         int n = s.length();
17         List<Integer> result = new ArrayList<>();
18
19         // Create doubled string to handle circular rotations
20         String doubled = s + s;
21
22         // For every rotation
23         for (int start = 0; start < n; start++) {
24
25             int max = 1;
26
27             // Check odd length palindromes
28             for (int center = start; center < start + n; center++) {
29
30                 int left = center;
31                 int right = center;
32
33                 while (left >= start &&
34                     right < start + n &&
35                     doubled.charAt(left) == doubled.charAt(right)) {
```

Line: 99 Col: 2

Upload Code as File

Test against custom input

Run Code

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Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

Sample Test case 0

Sample Test case 1

Sample Test case 2

Input (stdin)

Download

```
1 13
2 aaaaabbbbbaaaa
```

Your Output (stdout)

```
1 12
2 12
3 10
4 8
5 8
6 9
7 11
```